AI-Based Predictive Decision Support System

SparkIIT AI Internship Project 2 | Machine Learning, Predictive Analytics, Classification, Data Analysis

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Objective

To build a decision-support system that estimates the probability that a borrower will experience serious financial distress (90 or more days past due) within two years, compares several machine learning models, converts the probability into a cost-aware approve / review / decline recommendation, explains each score, audits the system for age-related differences, and delivers it as an interactive web application. The system supports a human decision maker and does not replace one.

Dataset

Kaggle competition **Give Me Some Credit** (https://www.kaggle.com/c/GiveMeSomeCredit), file *cs-training.csv*: 150,000 real borrowers, 10 inputs (credit utilization, age, three past-due counts, debt ratio, monthly income, open credit lines, real estate loans, dependents) and the target *SeriousDlqin2yrs*. After removing one row with an impossible age, **149,999** borrowers remained, of whom **6.68%** became seriously delinquent. A stratified 80/20 split gave **119,999 training** and **30,000 test** borrowers. Monthly income and number of dependents contain missing values, the past-due columns contain error codes (96 and 98), and credit utilization and debt ratio contain extreme outliers, so data cleaning was a real part of the work.

Key Work Done

- Checked data quality: missing values, impossible ages, special codes in the past-due columns and extreme utilization values.
- Split the data **before** computing any statistic. Caps for extreme values (99th percentile for utilization and debt ratio, 99.5th for income) come from the training part only.
- Engineered five features: total past-due events, an ever-late flag, log income, log debt ratio and income per dependent.
- Built one leak-free pipeline: median imputation with missing-value flags, scaling, ANOVA feature selection (number of features tuned) and the model.
- Compared Logistic Regression, Random Forest and Hist Gradient Boosting with 5-fold cross-validation using ROC-AUC, PR-AUC and Brier score. Plain accuracy was deliberately not used (see Analysis).
- Tuned the best model with 3-fold GridSearchCV. The selected Random Forest uses depth 12, minimum leaf size 20 and all features.
- Chose the decline threshold by minimizing an assumed cost (approving a borrower who later defaults is 5 times worse than declining a good one), then checked it on the untouched test set.
- Explained the model with permutation importance on the original inputs and added a per-borrower breakdown to the app.
- Audited decisions across three age groups, with and without age in the model.
- Delivered a custom Flask web application (HTML, CSS and JavaScript front end) with an input form, a recommendation shown on a risk scale, an explanation of the main drivers, and performance and fairness sections.

Results

Model	ROC-AUC	PR-AUC	Brier (lower is better)
Logistic Regression	0.8560 +/- 0.0069	0.3848 +/- 0.0092	0.0500 +/- 0.0004
Random Forest (selected)	0.8631 +/- 0.0061	0.3992 +/- 0.0083	0.0492 +/- 0.0004
Hist Gradient Boosting	0.8622 +/- 0.0056	0.3974 +/- 0.0079	0.0492 +/- 0.0004

Table 1. 5-fold cross-validation on the training set (mean +/- standard deviation across folds).

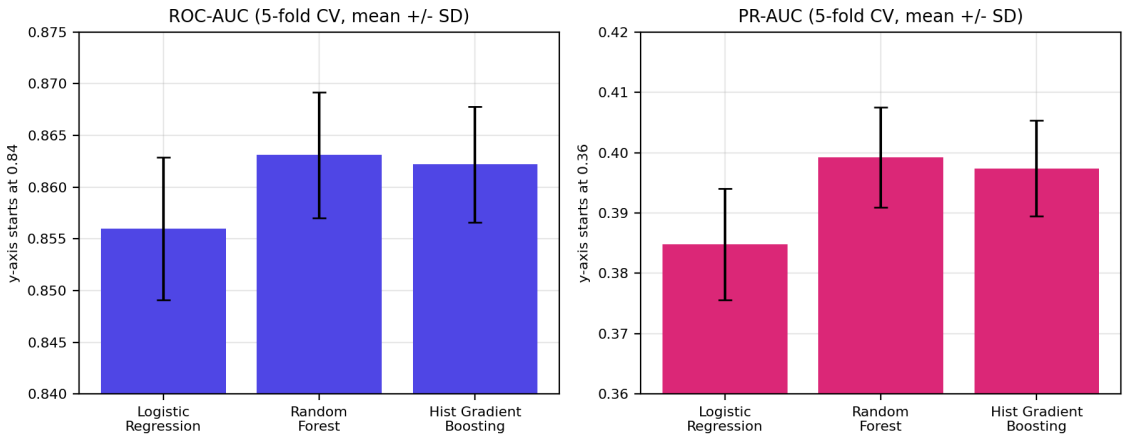


Figure 1. Cross-validated model comparison (error bars show the fold-to-fold standard deviation).

Held-out test result (30,000 borrowers)	Value	Reference point
ROC-AUC	0.8681	0.5 = random ranking
PR-AUC	0.3956	0.0668 for a model with no skill (the default rate)
Brier score	0.0490	0.0623 when everyone is given the average risk

Table 2. Test-set ranking quality of the tuned Random Forest.

Decision rule at the 16% decline threshold	Value
Borrowers declined	11.0%
Default rate among approved borrowers	3.1% (overall rate 6.7%)
Default rate among declined borrowers	35.4%
Share of all defaulters caught (recall)	58.2%
Precision of declines / F1 of declines	35.4% / 0.44
Average cost per applicant: approve everyone / decline everyone / this model	0.334 / 0.933 / 0.211

Table 3. What the cost-based decision rule achieves on unseen borrowers.

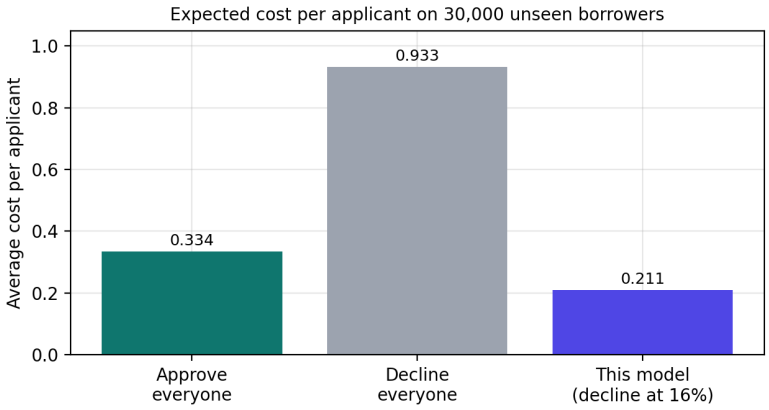


Figure 2. Expected cost per applicant under three policies (assumed 5:1 cost ratio).

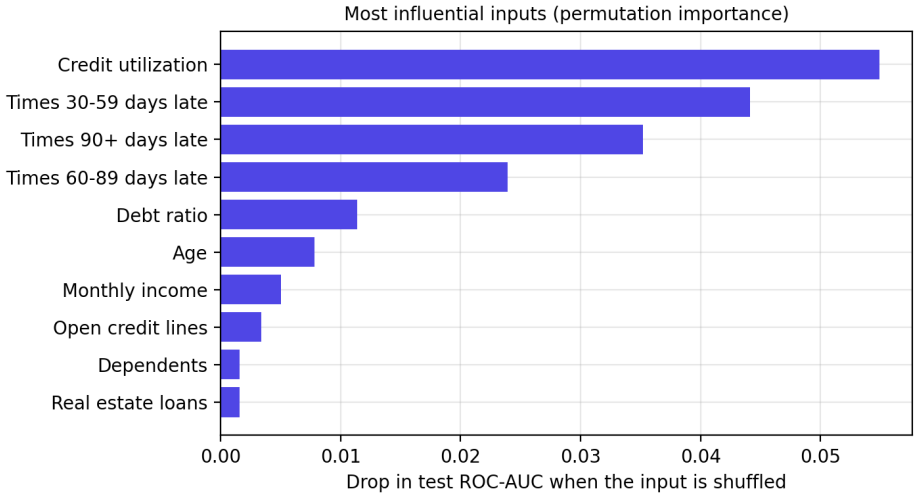


Figure 3. Permutation importance of the original inputs on the test set.

Analysis

- **Accuracy is the wrong yardstick here.** Only 6.7% of borrowers default, so approving everyone is already 93% accurate while catching no defaulter. ROC-AUC, PR-AUC, Brier score and the cost of decisions measure what matters.
- **Model comparison.** Random Forest (0.8631) and Hist Gradient Boosting (0.8622) differ by 0.0009, far less than the fold-to-fold spread of about 0.006, so they are effectively tied. Logistic Regression is about 0.007 lower, roughly one standard deviation. Tuning moved the cross-validated score from 0.8631 to 0.8639 (measured on different folds), which is not a meaningful change.
- **Generalization.** The test ROC-AUC of 0.8681 sits within about one standard deviation of the cross-validated 0.8631, so there is no sign of overfitting or leakage. The PR-AUC of 0.3956 is about six times the no-skill level of 0.0668, and the Brier score of 0.0490 is about 21% below the 0.0623 of a model that gives everyone the average risk.
- **What the decision rule does.** Declining the 11% of borrowers with the highest predicted risk catches 58% of all defaulters. Among borrowers who are approved, the default rate falls to 3.1% from 6.7%. Under the assumed 5:1 cost ratio, the expected cost per applicant is 0.211, which is 37% below the best policy without a model (approve everyone, 0.334). The threshold came from cross-validation on the training data and was then checked on test data it had never seen. The cost ratio itself is an assumption, and a real lender would use its own figures.
- **What drives the risk estimate.** Credit utilization and the three late-payment counts dominate. The late-payment columns are correlated, so their individual importances understate how much past-due history matters as a group. Age, income and debt ratio contribute modestly.

Fairness Audit (Age)

The dataset has no sex or race column, but age is present and is a sensitive attribute in lending. Decisions on the 30,000 test borrowers were compared across three age groups, first with age in the model and then with age removed.

Age group	n	Actual default	Mean predicted risk	Declined (with age)	Declined (no age)	Good borrowers declined (with)	Good borrowers declined (no age)	Defaulters caught (with)	Defaulters caught (no age)
under 30	1,803	12.6%	11.3%	19.6%	19.0%	13.1%	12.6%	64.3%	63.4%
30 to 59	18,546	8.1%	7.9%	13.4%	13.1%	9.4%	9.2%	58.9%	57.9%
60 and over	9,651	2.9%	3.4%	4.6%	5.1%	3.3%	3.8%	49.5%	50.9%

Table 4. Decisions by age group. Test ROC-AUC is 0.8681 with age and 0.8641 without age.

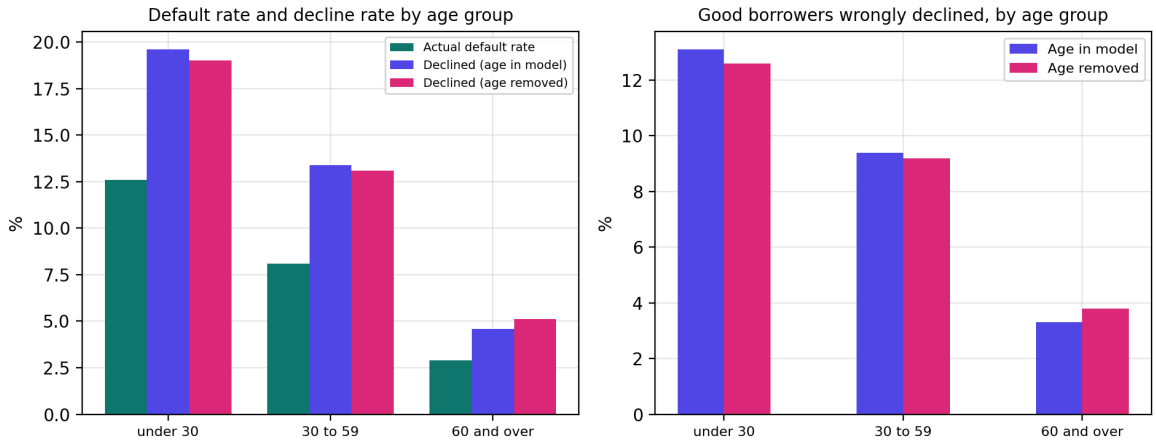


Figure 4. Default and decline rates by age group, with and without age in the model.

- **Predictions track reality on average in every group.** Mean predicted risk (11.3%, 7.9%, 3.4%) is close to the actual default rate (12.6%, 8.1%, 2.9%), so the model is not simply over-penalizing one group on average.
- **The burden of errors is uneven.** Declines range from 19.6% of borrowers under 30 to 4.6% of those aged 60 and over, which partly reflects real differences in default rates. But among borrowers who did not default, 13.1% of those under 30 were still declined versus 3.3% of those aged 60 and over, about four times as often.
- **Removing age does not fix it.** Without age the ROC-AUC drops by only 0.004 and decline rates barely move (19.0%, 13.1%, 5.1%). The likely reason is that other inputs, such as credit history and utilization, carry age-related signal and act as proxies. This explanation was not tested directly.
- **Caution.** The under-30 group has 1,803 borrowers, so its rates carry roughly one point of sampling noise. Which fairness criterion is acceptable is a policy and legal decision, and this audit is not a compliance review.

Limitations and Future Work

- The data is anonymized US consumer data from around 2011, so the model cannot be assumed to transfer to other countries or time periods. Validating on later data would test that.
- The target is serious delinquency, not profit. The 5:1 cost ratio is an assumption, and a sensitivity analysis over several ratios would show how the threshold moves.
- Final numbers come from one train/test split, although cross-validation agrees closely with it.
- Predicted probabilities could be recalibrated (for example with isotonic regression) and explanations could use SHAP values.
- Fairness-aware thresholds per group, and monitoring for drift after deployment, would be the next steps toward a production system.

Outcome

Delivered an end-to-end, explainable credit-risk decision-support system trained on 150,000 real borrowers. The tuned Random Forest reaches a test ROC-AUC of **0.868**. A cost-based rule declines 11% of applicants, catches 58% of eventual defaulters and lowers the expected cost per applicant by 37% against the best no-model policy under the stated assumption, while the default rate among approved borrowers falls to 3.1% from 6.7%. The project also documents how it was validated, where it is uncertain, and how decisions differ across age groups, and it ships as an interactive web application.